Why does token refresh fail on cold start? Check `@src/auth/session.ts` and run the tests.

Build sonnet-4.5

Thinking · 4s

read src/auth/session.ts 1–120

Two refreshes can start at once on cold start, so the second one invalidates the token the first just received. Running the auth tests to confirm.

bash running · 6s

```
$ npm test -- auth
PASS src/auth/login.test.ts
FAIL src/auth/session.test.ts
• refresh › concurrent calls
Expected 1 call, received 2
```

Message, / for commands, @ for files

Build | sonnet-4.5 | high

429-error.png
Pasted image · 184 KB

Users get a 429 with no retry hint. Add a limiter per API key.

Plan

1. Add `limiter.ts` with a token bucket
2. Send `Retry-After` on 429
3. Cover both paths with tests

edit src/middleware/limiter.ts +12 −4

```
@@ -18,7 +18,11 @@
export function limit(req, res, next) {
- if (hits[key] > MAX) return res.sendStatus(429)
+ const wait = bucket.take(key)
+ if (wait > 0) {
+ res.set('Retry-After', String(wait))
```

todos 2 of 4

- ✓ ~~Reproduce 429 without hint~~
- ✓ ~~Add token bucket~~
- ○ Send Retry-After header
- ○ Write tests

standup-summary.mp3 · 0:09 / 0:28

Message, / for commands, @ for files

+ Build sonnet-4.5 high

What does the release build type do differently from debug?

Commands matching /re — 4 results

BUILT-IN

/redo
Restore the last undone message and its file changes — Git repo

/sessions
Switch session · alias /resume

PROJECT

/review-changes
Review the last 10 commits — subagent

SKILLS

/pr-review
Review a pull request against team rules — skill

/re

+ | Build | sonnet-4.5 | ↑

q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123 | Done

edit src/auth/session.ts +12 -4

Fix is in. Committing and pushing the branch so CI can run the full suite.

bash · waiting for approval

Run a command?

Build agent wants to push to a remote

BASH

```
$ git push origin feature/refresh-token
```

in ~/projects/musa-api

"Always allow" would approve for this session

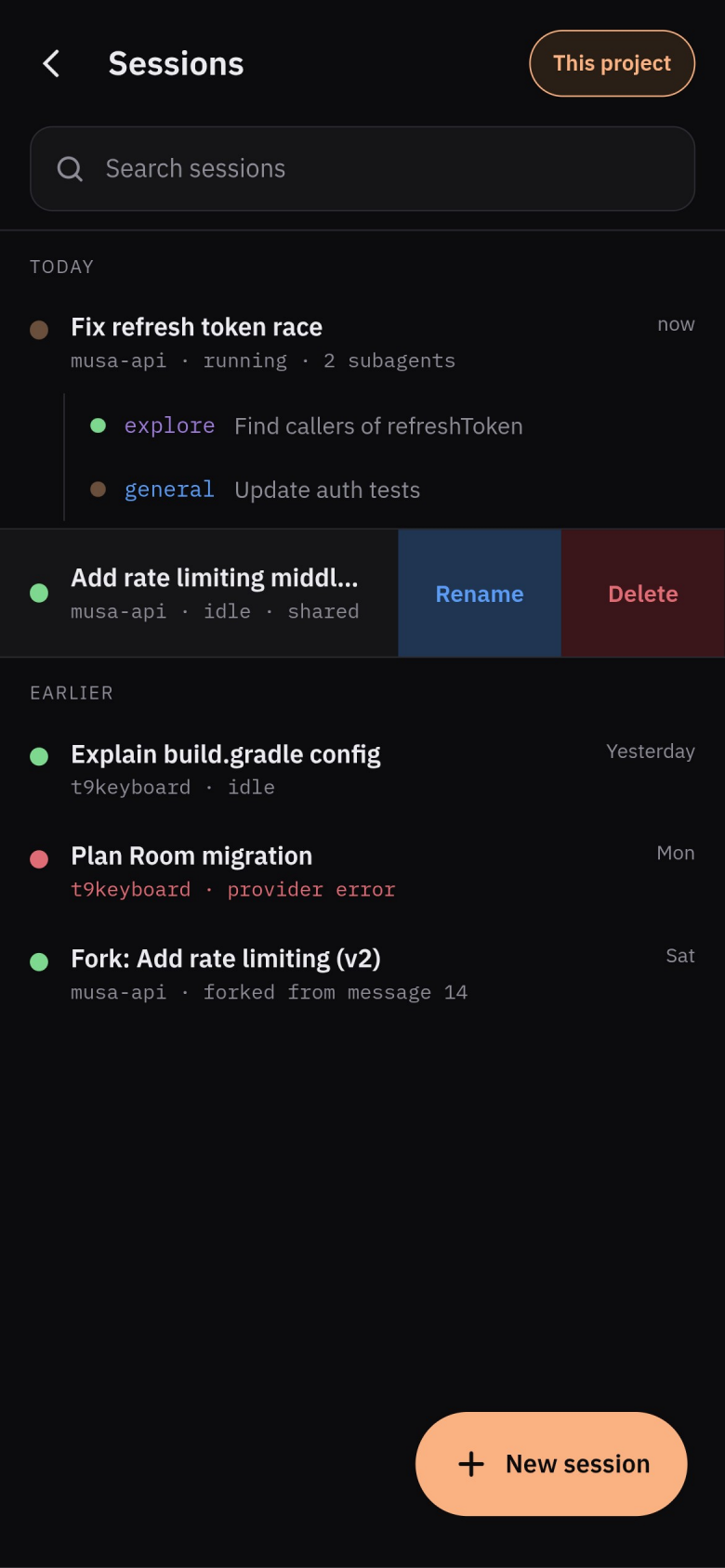
`git push*`

Auto-approve is off · 1 more request waiting

Allow once

Always allow

Reject


Sessions
This project
Search sessions
TODAY
Fix refresh token race
now
musa-api · running · 2 subagents
explore Find callers of refreshToken
general Update auth tests
Add rate limiting middl...
musa-api · idle · shared
Rename
Delete
EARLIER
Explain build.gradle config
Yesterday
t9keyboard · idle
Plan Room migration
Mon
t9keyboard · provider error
Fork: Add rate limiting (v2)
Sat
musa-api · forked from message 14
New session

Agents

Primary | **Subagents** | System

general built-in — inherits model

Multi-step research and parallel work. Full tools except todo.

edit bash web

explore built-in — inherits model

Fast, read-only codebase search. Cannot modify files.

read-only

scout experimental

Reads external docs and dependency source. Read-only.

code-reviewer custom — sonnet-4.5

Reviews code for security, performance and maintainability.

edit denied bash: ask

docs-writer custom hidden

Writes and maintains project documentation. No bash.

+ New agent

`@explore` find every caller of refreshToken, and update the auth tests in parallel.

I'll split this into two subagents. The search is read-only, so it runs alongside the test edits.

explore
done · 14s · 6 steps
Find all callers of refreshToken

7 call sites in 4 files. Only `api/client.ts` calls it without a lock.

Open session >

general
running · 31s · 9 steps
Update auth tests for the new mutex

`edit src/auth/session.test.ts`

Open session | Stop

Finished · 14s | Read-only agent | Depth 1 of 1

TASK FROM BUILD

Find all callers of refreshToken and note any that don't hold the session lock.

✓ grep refreshToken — 7 matches

```
src/auth/session.ts:42
src/auth/session.ts:77
src/api/client.ts:31
src/api/retry.ts:12
+ 3 more
```

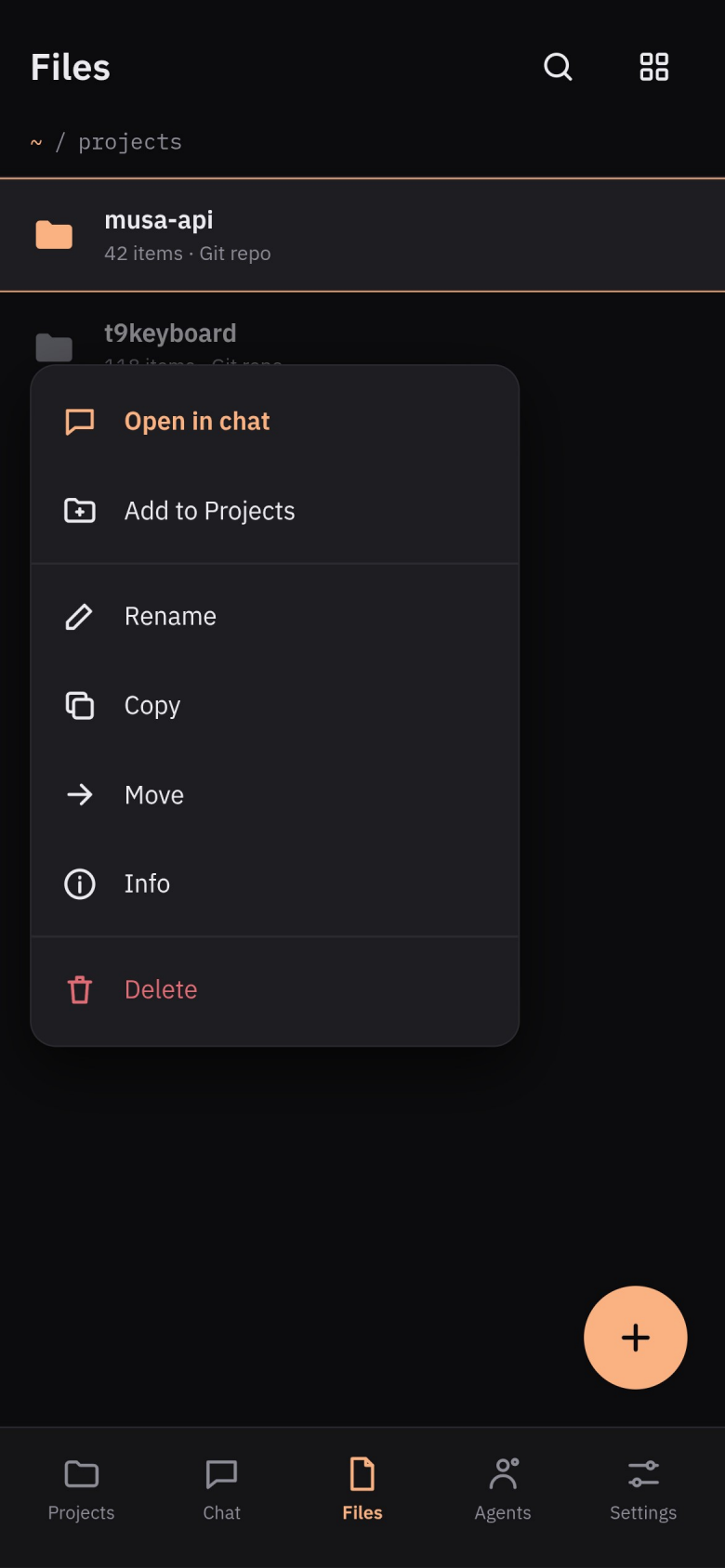
✓ read src/api/client.ts — 20–60

Result

Seven call sites across four files. Six go through `withLock()`. The one in `api/client.ts` line 31 does not, which matches the race.

↑ Return to parent

Swipe left or right to switch between subagent sessions


Files
~ / projects
musa-api
42 items · Git repo
t9keyboard
Open in chat
Add to Projects
Rename
Copy
Move
Info
Delete
Projects
Chat
Files
Agents
Settings

Settings

OpenCode server running
v2.0.6 · 127.0.0.1:4096 · Alpine

Restart

MODELS

- Providers & models — sonnet-4.5
- Agents & subagents — depth 1

SAFETY

- Permissions — ask for bash
- MCP servers — 3 connected
- Skills & commands — 9 available

APP

- Appearance — opencode · dark
- Notifications — on
- Sessions & data — share: manual
- Server & runtime — 212 MB